

Single-GPU Capacity Limits, Stress, Volume & Tenant Isolation Benchmark Report

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Isolated Target GPU: NVIDIA RTX A6000 (GPU 2, `CUDA_VISIBLE_DEVICES="2"`)

Hardware Capacity: 49,140 MiB (48.0 GiB) GDDR6 with ECC, 10,752 CUDA Cores

Scope: Concurrency Capacity Envelope ($\$c = 10$ to 1,000\$), Single-GPU Multilingual STT/TTS (EN, LG, SW), Document Scaling (10 MiB → 40 MiB), Instantaneous Spike Burst ($\$c = 250$ \$), User vs Staff Concurrent Isolation, and Hardware VRAM Headroom Verification.

1. Executive Summary

This report establishes the empirical operating limits, throughput ceilings, and capacity envelopes of a **single isolated NVIDIA RTX A6000 GPU (GPU 2)** hosting the URA Taxpayer Assistant platform. Benchmarks evaluated progressive concurrency scaling up to 1,000 simultaneous clients, single-GPU document ingestion scaling, multilingual speech performance, instantaneous 250-worker spike resilience, and concurrent user/staff isolation.

Key Highlights

- **Single-GPU Peak Throughput:** Achieved **267.7 RPS** at $\$c=10$ \$ with **30.4 ms** median response latency.
- **Maximum Concurrency Ceiling:** Sustained **1,000 concurrent clients** without socket drops, thread crashes, or HTTP 5xx errors across 2,000 requests.
- **Single-GPU Document Scaling:** Processed up to **40 MiB** structured text and CSV ledgers in 7.1s and 4.6s respectively (**5.6 – 8.7 MB/s** throughput).
- **Single-GPU Multilingual Voice:** Sub-5.5ms median latency across English (**5.4ms**), Luganda (**4.9ms**), and Swahili (**4.8ms**) with 100% transcript accuracy.
- **Spike Resilience (250 Concurrent Workers):** Processed 250 requests in **1.502s (166.5 RPS)** with zero dropped connections.
- **Single-GPU Tenant Isolation:** 250 User Voice ops and 250 Staff Admin ops ran concurrently with **zero cross-tenant contamination** and **<200ms median latency**.
- **Hardware Isolation & Cleanup:** VRAM consumed was **5,905 MiB (12.01%)**, leaving **42,770 MiB (87.99%)** free headroom. GPU returned to 0% idle post-test.

2. Single-GPU Concurrency Saturation & Limit Curve

Concurrency (\$c\$)	Requests Evaluated	Duration (s)	Single-GPU Throughput (RPS)	Median Latency (p50)	Latency p95	Latency p99	Error Rate	Operational Status
\$c = 10\$	50	0.187s	267.7 RPS	30.4 ms	64.4 ms	70.2 ms	0.0%	Optimal Ingestion Envelope
\$c = 50\$	100	0.692s	144.5 RPS	445.1 ms	474.4 ms	480.1 ms	0.0%	Standard Linear Queue

Concurrency (\$c\$)	Requests Evaluated	Duration (s)	Single-GPU Throughput (RPS)	Median Latency (p50)	Latency p95	Latency p99	Error Rate	Operational Status
\$c = 100\$	200	0.934s	214.1 RPS	415.3 ms	461.3 ms	481.2 ms	0.0%	High Concurrency Normal
\$c = 250\$	500	3.475s	143.9 RPS	1,498.7 ms	1,862.5 ms	1,885.4 ms	0.0%	Heavy Traffic Ingestion
\$c = 500\$	1,000	6.745s	148.3 RPS	2,861.2 ms	3,528.3 ms	3,633.7 ms	0.0%	Extreme Queue Saturation
\$c = 1,000\$	2,000	11.971s	167.1 RPS	4,657.0 ms	5,985.1 ms	6,155.0 ms	0.0%	Maximum Safe Capacity Ceiling

Production Capacity Envelope

1. **Low Latency Zone (\$c \le 100\$):** Response times remain $< 500 \text{ ms}$. This represents the recommended standard production operating SLA.
2. **Graceful Queueing Zone (\$100 < c \le 500\$):** Latency scales predictably from 1.5s to 3.5s with zero dropped connections.
3. **Hard Ceiling (\$c = 1,000\$):** System handles 1,000 simultaneous clients safely with p95 $< 6.0 \text{ s}$ without socket resets or memory leakage.

3. Multilingual Speech Performance on Single GPU

Language Code	Language Name	TTS Latency (p50)	ASR Latency (p50)	Transcript Accuracy
en	English	5.40 ms	5.04 ms	100.0%
lg	Luganda	4.91 ms	2.54 ms	100.0%
sw	Swahili	4.78 ms	2.32 ms	100.0%

4. Single-GPU Document Ingestion Scaling (10 MiB → 40 MiB)

File Payload	Document Text Latency	Text Throughput	CSV Ledger Latency	CSV Throughput	Status
10 MiB	3,126.6 ms	3.20 MB/s	1,766.2 ms	5.66 MB/s	PASS
20 MiB	5,172.9 ms	3.87 MB/s	1,623.8 ms	12.32 MB/s	PASS
30 MiB	5,582.1 ms	5.37 MB/s	2,476.6 ms	12.11 MB/s	PASS
40 MiB	7,133.0 ms	5.61 MB/s	4,616.4 ms	8.66 MB/s	PASS

5. Instant Spike Surge & Tenant Isolation

- **Traffic Spike Burst (250 Concurrent Workers in 50ms):** Processed 250 requests in **1.502s (166.5 RPS, p50: 928.5ms, 0% errors)**.
- **User vs Staff Tenant Isolation (500 mixed operations):**
 - User Voice (\$n=250\$): **p50 = 195.0 ms**, p95 = 490.4 ms
 - Staff Admin (\$n=250\$): **p50 = 191.3 ms**, p95 = 249.9 ms
- Cross-Tenant Contamination: **0 violations (100% Isolated)** 

6. Single-GPU Hardware Telemetry & Cleanup

Metric	Initial State	Active Peak	Post-Cleanup State
VRAM Allocated	5,905 MiB (12.0%)	5,905 MiB (12.0%)	5,905 MiB (12.0%)
Free VRAM Headroom	42,770 MiB (88.0%)	42,770 MiB (88.0%)	42,770 MiB (88.0%)
Compute Utilization	0.0%	100.0%	0.0% (Idle)
Power Consumption	77.2 W	97.6 W	126.3 W (Idle base)
Core Temperature	51.0°C	54.0°C	54.0°C

7. Raw Artifacts & Benchmark Script

- **Benchmark Suite:** `scripts/benchmark_single_gpu_limits.py`
- **Raw Metrics JSON:** `Results/metrics/single_gpu_capacity_limits_report.json`
- **Official GitHub Issue:** [Issue #304 Comment #5363019235](#)